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Source: *Biometrics*, Vol. 51, No. 4 (Dec., 1995), pp. 1197-1205
Published by: The International Biometric Society
Stable URL: https://www.jstor.org/stable/2533253
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BIOMETRICS 51, 1197–1205
December 1995

Testing for the Presence of Immune or Cured Individuals in Censored Survival Data

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SUMMARY

In a sample of censored survival times, the presence of an immune proportion of individuals who are not subject to death, failure, or relapse, etc., may be indicated by a relatively high number of individuals with large censored survival times. We summarise some recent theoretical work which justifies analogues of the usual model fitting and testing techniques for such data. In particular, we discuss a 'boundary' test for the presence of immunes in the population and goodness of fit tests for parametric descriptions of the data. The methods are illustrated on some data on the relapse times of leukemia patients.

1. Introduction

Models which allow a component of 'immune' or 'cured' individuals to be present in a population otherwise subject to death or failure have been fitted to survival data since the time of Boag (1949), but their widespread acceptance by biostatisticians and epidemiologists seems to have lagged, as discussed by Chen et al. (1985). This may be due in part to a relative dearth of clearly defined methodological and computational results with which to place these models in the mainstream of modern statistical practice. Notwithstanding this, there has been a groundswell of interest in the application of these kinds of models; see for example Goldman (1984), Sposto, Sather, and Baker (1992), Cantor and Shuster (1992), and their references.

In this paper we summarise some recent work which fills the above-mentioned gaps to a large extent, and can be applied to justify the kinds of procedures we are used to employing in survival analysis, such as testing for the effects of covariates by fitting a parametric distribution like the exponential, gamma, or Weibull, etc. 'Mixture' models allowing for immunes can be fitted to data, and data analysis can be performed, in an exactly analogous way to these models, and with little extra computational difficulty, provided the survival times are approximately drawn from a parametric distribution such as an exponential or Weibull distribution. Furthermore, they are amenable to rigorous likelihood analysis of precisely the type applied to generalised linear models (GLMs) and the Cox proportional hazards model, for example. An important aspect of analyses such as the above is to consider whether or not an immune proportion does in fact exist in the population. Under the parametrisations commonly proposed, this requires testing at the boundary of the parameter space, and the main aim of the paper is to discuss and exemplify such a test. The test we propose is based on the log-likelihood ratio statistic, and its asymptotic distribution is 'non-standard', i.e., it is not asymptotically distributed as a chi-squared random variable. We illustrate its application on a set of leukemia data, and also demonstrate a method for assessing the goodness of fit of the parametric distribution (the exponential) used to model the censored data.

To specify the problem, consider a population consisting of two components. The first, which is present in proportion p, say, contains 'susceptibles' who are subject to an event such as death, relapse, or failure, etc. The 'lifetime' (time to occurrence of the event) of such individuals is assumed to follow an exponential distribution. These individuals are subject to right censoring. The remaining proportion $1 - p$ of the population is 'immune to' or 'cured of' the disease, and for these individuals the event of interest never occurs. Thus observations on their lifetimes are always censored at the limit of follow up. We cannot identify the immune individuals if any, in a sample, but

Key words: Immune proportion; Survival data; Likelihood ratio test.

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their presence will be signaled by a leveling of the Kaplan–Meier (1958) empirical distribution function estimator (KME) of the data at its right extreme (see for example, Figure 1 below).

As in Farewell (1982), we can model this situation by assuming that an individual has a probability p of being subject to failure and $1 - p$ of being immune. Define an indicator random variable b by

$$b = \begin{cases} 1 & \text{if the individual is subject to failure;} \\ 0 & \text{if the individual is immune,} \end{cases}$$

with $P(b = 1) = p = 1 - P(b = 0)$. Let $t^* \geq 0$ be the survival time of an individual and suppose $P(t^* \leq t|b = 1) = F_0(t)$ for $t \geq 0$, where $F_0(t)$ is a cumulative distribution function (c.d.f.) which is 'proper', i.e., $F_0(\infty) = 1$. Then the (unconditional) distribution function of t^* can be expressed as

$$F(t) = P(t^* \leq t) = P(t^* \leq t|b = 1)P(b = 1) = pF_0(t). \tag{1.1}$$

When $p < 1$, $F(t)$ is 'improper' in that it has total mass less than 1 on $[0,\infty)$. We will take $F_0(t)$ throughout to be $1 - e^{-\lambda t}$, the exponential c.d.f. Then (1.1) becomes

$$F(t) = p(1 - e^{-\lambda t}). \tag{1.2}$$

Finally, we assume that lifetimes are subject to censoring, which as usual we assume to be noninformative.

An important question is whether there really is an immune proportion in the population, i.e., is the extra flexibility of the parameter p in model (1.1) really needed in our data, or may it be taken as 1? In the latter case we would have an 'ordinary' survival model with all individuals subject to failure, and analysis can proceed in the usual way.

The question of whether we may take p to be 1 or not involves testing at the boundary of the parameter space $(0,1]$ of p. This gives rise to a log-likelihood ratio test whose asymptotic distribution is not the chi-square distribution, but is a 50–50 mixture of a chi-square random variable with 1 degree of freedom and a point mass at 0 (see Zhou and Maller (1995) and our Appendix). We illustrate this theory by the analysis of a particular set of medical data, in which we show how to test for the presence of immunes in two groups of patients. Examination of this data suggests a problem of goodness of fit of the parametric exponential distribution to one of the groups, and we confirm this by proposing a goodness-of-fit test which is a variant of the Filliben test designed to assess goodness of fit of the normal distribution. In the absence of a parametric model for this group, we discuss a non-parametric analysis. We conclude with a brief discussion of some other issues arising in the analysis of this kind of data.

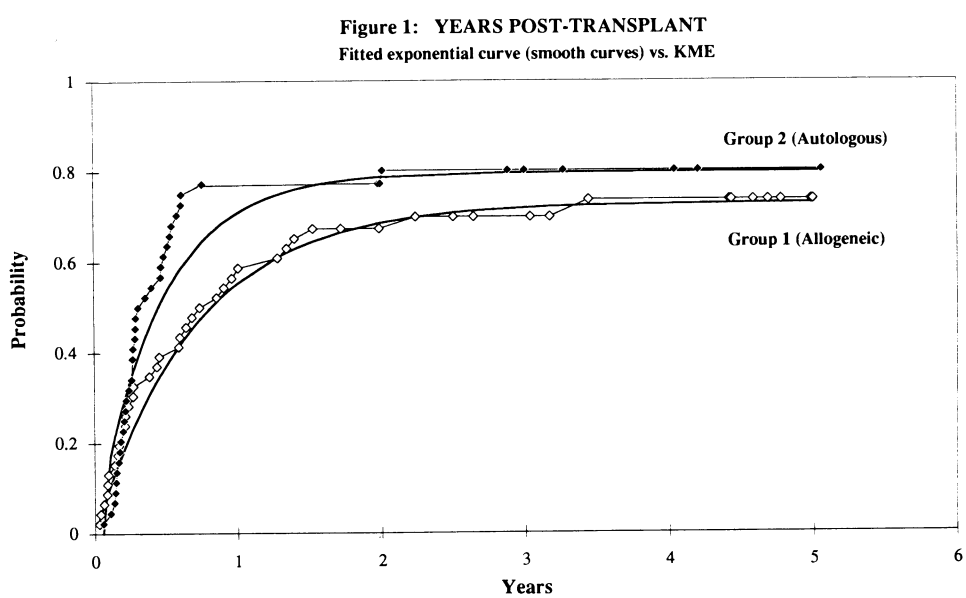

Figure 1. Kaplan—Meier empirical distributions, with fitted exponential mixture distributions (full lines), for Group 1 (allogeneic) and Group 2 (autologous). The fit to Group 1 is judged to be good, while that to Group 2 is poor (see Section 3).

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2. Testing for the Presence of Immunes

Figure 1 shows the Kaplan–Meier estimators calculated from some data, kindly supplied by Prof. Kersey and Dr. Goldman, on relapse times of leukemia patients following two types of transplant, allogeneic (Group 1) or autologous (Group 2) (see Kersey et al. (1987), especially Figure 5). (We will refer to the time till relapse as the 'survival time' of the patient.) The Figure shows each individual survival or censored time (accurate to within 1 day); censored times of course are those at which the KME does not increase. Group 1 consists of observations on 46 patients, of whom 13 were censored, while Group 2 consists of observations on 44 patients, of whom nine were censored (two censored observations are tied).

First let us concentrate on Group 1 (allogeneic transplants), for which the KME and a fitted exponential mixture model as in (1.2) are plotted. The parameter estimates (with 95% confidence intervals in brackets) obtained by maximising the likelihood were found to be

$$\tilde{p} = .73\ (.59,\ .86) \quad \text{and} \quad \tilde{\lambda} = .12\ (.07,\ .18). \tag{2.1}$$

(Our method of obtaining these estimates and their standard errors is briefly discussed at the end of Section 3; survival time is measured in 30-day periods or 'months'). Equation (2.1) reveals that the estimated proportion of immune patients for the allogeneic group is $1 - .73 = .27$. (In their context, Kersey et al. (1987) refer to these individuals as being 'cured', but we will retain the terminology 'immune'). The median relapse time for susceptibles in Group 1 is estimated to be

$$\frac{\log 2}{\tilde{\lambda}} = \frac{\log 2}{.12} = 5.8 \text{ months}. \tag{2.2}$$

Judging by the confidence interval for p in (2.1), there appears to be a proportion of immunes in the population, but we wish to test rigorously for this. Equivalently, we test as a null hypothesis

$$H_0{:}p = 1. \tag{2.3}$$

In general, denote by $\tilde{\theta} = (\tilde{p}, \tilde{\lambda})$ the maximum likelihood estimates (MLEs) obtained from fitting model (1.1) (taking $\tilde{p} = 1$ if the likelihood is maximised on the boundary) and let $\tilde{\theta}_{H_0}$ be the corresponding MLE when H_0 holds. Let $\ell_n(\theta)$ be the log-likelihood function and let

$$d_n = -2\{\ell_n(\tilde{\theta}_{H_0}) - \ell_n(\tilde{\theta})\} \tag{2.4}$$

be the 'deviance' statistic designed to test H_0. Then d_n has, asymptotically, a distribution which is a 50–50 mixture of a chi-square random variable (r.v.) with 1 degree of freedom (χ_1^2) and a mass at 0. (The same distribution is mentioned in Cox and Hinkley (1974, p. 321) in a different but related boundary testing problem.) We write this result as

$$P\{d_n \leqslant x\} \to \frac{1}{2} P\{\chi_1^2 \leqslant x\} + \frac{1}{2} \tag{2.5}$$

as $n \to \infty$, for $x \geqslant 0$. A proof of (2.5), for the case when λ is known, is sketched in our Appendix, and a general proof is given in Zhou and Maller (1995), under very mild restrictions on the censoring mechanism. An independently and identically distributed (i.i.d.) setup is assumed, wherein the survival times are observations on $t_i = \min(t_i^*, u_i)$, t_i^* being i.i.d. with exponential distribution $F_0(t) = 1 - e^{-\lambda t}$, and u_i being i.i.d. with distribution G, say, not degenerate at 0. The random variables t_i^* and u_i are assumed to be independent, and for each i, $E(e^{(\lambda+\delta)u_i})$ is assumed to be finite for some $\delta > 0$. This moment condition on u_i stipulates in effect that the tail of the censoring distribution G must decrease faster than that of F. (In the Appendix, we only need $E(e^{\lambda u_i})$ finite.)

Figure 1 and the goodness-of-fit test in Section 3 below suggest that the exponential distribution is a good fit to the Group 1 data. The value of $-2\ell_n(\tilde{\theta})$, with $\tilde{\theta}$ given by (2.1) for Group 1 data, is 257.4. To test H_0 we compare this value with 280.3, which is the maximised value of minus twice the log-likelihood for an ordinary ($p = 1$) exponential distribution fitted to the data. (This model was fitted to the censored survival times using GLIM; see Aitkin et al. (1989) and Baker and Nelder (1978). Note that the GLIM 'deviance' from the fitted model is not the value of $-2\ell_n(\tilde{\theta}_{H_0})$ that we require; we calculated this directly).

To make the test we calculate

$$280.3 - 257.4 = 22.9,$$

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which should approximately be an observation from the distribution in (2.5) if H_0 is true. The 95th percentile $c_{.95}$ of this distribution can be calculated from:

$$\frac{1}{2} + \frac{1}{2} P\{\chi_1^2 \leq c_{.95}\} = .95,$$

so $P\{\chi_1^2 \leq c_{.95}\} = .90$, giving

$$c_{.95} = 2.71.$$

Since $22.9 > 2.71$, we reject H_0 and consider that there is strong evidence for immunes in Group 1.

Note that the use of the asymptotic distribution (2.5) effectively increases the power of the test over what it would have been if we had erroneously taken d_n as χ_1^2. Goldman (1984) does this, but her results are easily corrected—we simply need to re-interpret her level α percentage points as level $\alpha/2$ points.

3. Testing for Goodness of Fit

Section 2 illustrates the boundary testing problem for Group 1. Now we discuss the analysis of Group 2 (autologous transplants). An exponential fit to the Group 2 data is poor, as can be judged from Figure 1. In fact it appears that none of the usual parametric survival distributions would well fit this data, with its almost uniform (and fast!) initial failure rate.

In this Section we propose a method of testing the goodness of fit of a parametric distribution to censored survival data which is akin to Filliben's (1975) test for normality of uncensored data. Filliben suggested calculating the correlation coefficient between the order statistics of the sample and the expected values (or medians) of the order statistics for a sample of the same size from the standard normal distribution.

Our suggested test is a variant of this. Consider the hypothesis

$$H_0: F = \tilde{F}$$

where $\tilde{F}$ is a specified c.d.f. such as the exponential. Let $t_{(1)} \leq t_{(2)} \leq \cdots \leq t_{(n)}$ be an ordered sample of survival times. Suppose for the moment that there is no censoring. A '*P-P*' (probability) plot of the data is a plot of $\tilde{F}(t_{(i)})$ against $i/(n+1)$, $1 \leq i \leq n$. This should be close to a straight line if H_0 is true, the rationale being that when $F = \tilde{F}$, $\tilde{F}(t_i)$ has the uniform distribution over [0,1) ($U[0,1]$), so $\tilde{F}(t_{(i)})$ has the same expectation as that of the ith order statistic in a sample of n from $U[0,1)$, which is $i/(n+1)$.

Now let $\hat{F}_n$ be the empirical distribution function of the sample $t_1, \ldots, t_n$. Since $\hat{F}_n(t_{(i)}) = i/n$, the *P-P* plot is equivalent to plotting $\tilde{F}(t_{(i)})$ against $\hat{F}_n(t_{(i)})$. When the data are subject to censoring, the same argument leads us to expect to obtain a near-straight line with slope close to 1 by plotting $\tilde{F}(t_{(i)})$ against $\hat{F}_n(t_{(i)})$, when H_0 is true, provided we take $\hat{F}_n$ as the KME. When $\tilde{F}(t_{(n)})$ is close to $\hat{F}_n(t_{(n)})$, as it is in our case, the correlation coefficient r between $\tilde{F}(t_{(i)})$ and $\hat{F}_n(t_{(i)})$, $1 \leq i \leq n$, should provide an appropriate measure of the goodness of fit. Thus we propose to use this correlation coefficient to test the goodness of fit. Note that we include the censored observations in calculating the correlation.

Applying this calculation to the data in Section 2, we find for Group 1 (allogeneic) a value of $r = .9959$ for a fit of the improper exponential

$$\tilde{F}(t) = \tilde{p}(1 - e^{-\tilde{\lambda} t}) \tag{3.1}$$

with $\tilde{p}$ and $\tilde{\lambda}$ as given by (2.1). On the other hand, for Group 2 (autologous) we find that $r = .9709$ for the comparison of $\hat{F}_n(t)$ with $\tilde{F}(t)$, where $\tilde{F}(t)$ is given by (3.1) with the following MLEs:

$$\tilde{p} = .83 \quad \text{and} \quad \tilde{\lambda} = .21. \tag{3.2}$$

Figure 2 shows $\tilde{F}(t_{(i)})$ plotted against $\hat{F}_n(t_{(i)})$, for each group.

In accordance with our intuition, the values of r suggest a good fit for Group 1 and a poor fit for Group 2, and now we wish to test rigorously for these effects. In the case of uncensored data, it was relatively straightforward for Filliben (1975) to simulate and tabulate the percentage points of the distribution of his test statistic. When survival times are censored, however, we must consider the effect of the unknown censoring distribution on the null distribution of r. In the absence of theoretical guidance we proceed by doing some simulations for a range of censoring distributions and tabulating the 5th and 80th percentiles of the simulated distributions of r.

Common censoring distributions considered in literature are usually the exponential or the

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Figure 2: P-P plots

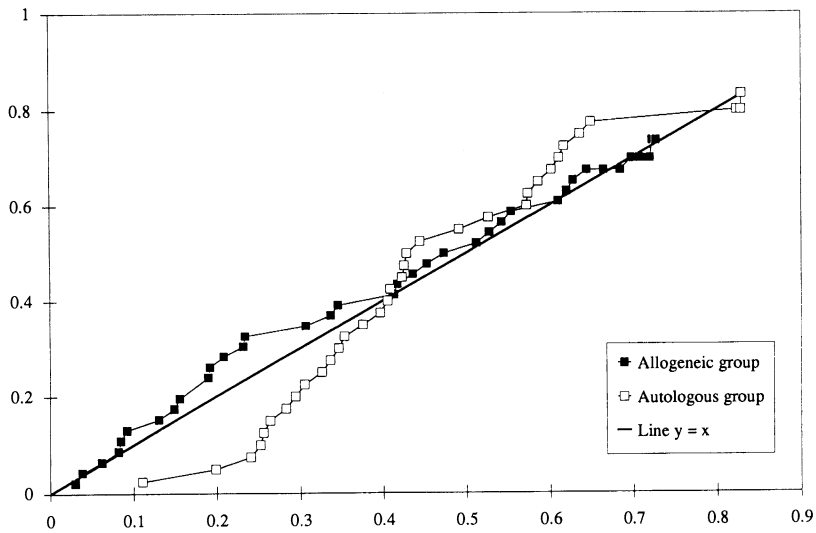

Figure 2. Plot of the KME $\hat{F}_n(t_i)$ versus $\tilde{F}(t_i)$, where t_i are survival times and $\tilde{F}$ is the exponential c.d.f. with parameters given by (2.1). Group 1 (allogeneic) is represented by dark squares, Group 2 (autologous) is represented by light squares. The straight line is the line **y** = **x**.

uniform, and we restrict ourselves here to these two. Since they are quite different in shape they cover a relatively wide range. Of course we could easily add to the repertoire by doing more simulations. Thus for Table 1 we took survival times from an improper exponential distribution with mean 1 (we can always transform the exponential to this situation) and values of p of .4, .6, .8, and 1. Censoring distributions were exponential with mean $\mu = 1$ or 2, corresponding to heavy and light censoring, or uniform with mean of 1 or 2. We took 1,000 repetitions of samples of size $n = 50$ of such censored survival times, and for each repetition calculated the Kaplan–Meier estimator $\hat{F}_n(t)$, then the correlation coefficient r of $\hat{F}_n(t_i)$ with the corresponding values of the underlying survival distribution

$$\tilde{F}(t_i) = p(1 - e^{-t_i/\mu}) \qquad (t \geqslant 0).$$

Table 1 shows the 5th and 80th percentiles of the distribution of r as estimated from the 1,000 repetitions. The table, although limited, suggests that light censoring or large values of p tend to produce larger correlation coefficients than heavy censoring or small values of p, as we might intuitively expect. Moreover, for a given immune proportion, the form of the censoring distribution does not seem to have much impact on the r values, as the percentiles do not differ too much for censoring distributions as different as the exponential and the uniform. The percentiles of the exponential distributions are generally smaller than those of uniform distributions with the same

Table 1
Eightieth and fifth percentiles of correlation coefficient

		Censoring distribution				
			exp(1/μ) With			$U[0, B]$ With
	p	μ:	1	2	B:	2	4
80%	0.4		.973	.984		.980	.987
	0.6		.982	.989		.986	.991
	0.8		.986	.992		.989	.993
	1		.990	.994		.991	.994
5%	0.4		.848	.914		.874	.926
	0.6		.904	.942		.925	.956
	0.8		.933	.966		.948	.969
	1		.957	.974		.964	.975

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means, and this is due to the skewness of the exponential distribution toward 0, corresponding to heavier censoring, and consistent with the above observation.

Table 1 further suggests that, for sample sizes of order 50 from an improper exponential survival distribution with $p = .8$, values of r exceeding .993 indicate a good fit. (Note that in common practice, a correlation coefficient greater than the 10th percentile is considered as being indicative of a reasonable fit, while .993 in fact is the 80th percentile). The value of $r = .9959$ for the allogeneic group thus lends support to our visual deduction of good fit in this group.

On the other hand, however, the value of $r = .9709$ for the autologous group is close to the 5th percentile of .969 for $p = .8$ and a light uniform ($U[0,4]$) censoring, though it is not conclusive in rejecting the exponential fitting. But notice that the censoring in the data for both groups is in fact very light—there are no censored observations before 18 months, whereas only a small proportion of individuals have survived beyond that time. Therefore it appears appropriate to simulate r values for samples which more closely resemble our data. Consequently, we did two more simulations of 1,000 repetitions, drawing survival times from improper exponential distributions with parameters given by (2.1) and with $n = 46$ for Group 1 (allogeneic), and by (3.2) and with $n = 44$ for Group 2 (autologous). The censoring distribution was chosen as $U[1.5,5]$ (in years) for both groups. For Group 1, the simulated 80th percentile of the null distribution of r was .9952, which, compared with an observed value of .9959, confirms the good fit of the exponential distribution to this group. For Group 2, the simulated 5th percentile of the null distribution of r was .9766, well above the observed value of .9709 for the group (in fact the value of .9709 is slightly less than the 2nd percentile of the simulated r values). On this basis we reject the exponential fit to the autologous data at a 5% (or even 2%) level of significance.

The exponential mixture models were fitted on a Fortran program using a simple Newton–Raphson algorithm to find a zero of the first derivative of the log-likelihood, and thus the parameter estimates in (2.1) and (3.2). Standard errors for the estimates were obtained from the inverse of the negative second derivative matrix evaluated at the MLE, and these were used to calculate the confidence intervals in (2.1). Simulations were done using the standard Fortran random number generator.

4. Discussion

When a parametric model such as the exponential is a good fit to the data we can test for the effects of covariates, such as the difference between the allogeneic and autologous groups, using a covariate method such as that of Maller (1993). But in Section 3 we confirmed the lack of fit of the exponential to the autologous data, and it is doubtful, by inspection, whether any of the ordinary parametric models such as the gamma or lognormal, etc., would better fit this data. In fact we also fitted the Weibull distribution to each group using a program described in Maller (1993). For Group 1, 2 ln only changed by an insignificant amount (.2), so the exponential is just as good a description of this set of data as the Weibull. However the Weibull was also a poor fit to the Group 2 data. Although it is not the main purpose of this paper, it seems worthwhile to discuss alternative analyses, with a view to testing for differences between the two groups in practice.

Another 'semi-parametric' approach to this data would be via the piecewise exponential model with covariates (e.g., Aitkin et al. (1989, p. 287)), extended by the inclusion of parameters to allow for immune or cured proportions in the groups. The data would lend itself quite well to this approach in that the hazard appears to be approximately constant in a few well defined intervals. At present, however, we do not have the practical or theoretical tools to handle such a model.

We could also consider the possibility that the lack of fit revealed in Figure 1 may be due to the presence of further exponentially distributed components in the data. These may be caused by the presence of subgroups with differing survival rates whose effect could be allowed for by the introduction of further covariates or factors into the model. The values of these would have to be obtained from the experimenter and are not available to us. So we turn now to non-parametric approaches.

We can test non-parametrically for differences in immune proportions as follows. Maller and Zhou (1992) and Sposto et al. (1992) proposed as an estimator of p the maximum observed value of the KME. In terms of Figure 1, these estimators, $\hat{p}_1$ (for allogeneic) and $\hat{p}_2$ (for autologous), say, are simply the values of the KME at its right extreme, i.e.,

$$\hat{p}_1 = .74 \quad \text{and} \quad \hat{p}_2 = .80.$$

Under an independent censoring model, Maller and Zhou (1992) showed that these estimators have some good properties provided follow up has been sufficient. In fact, $\hat{p}_j$ is a consistent and asymptotically normal estimator of the true proportion of immunes in group j, $j = 1, 2$, and each has

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estimated variance given by the appropriate version of $\hat{\sigma}^2$ as defined in Theorem 3 of Maller and Zhou (1992). For the given data these take the values

$$\hat{\sigma}_1^2 = .0181 \quad \text{and} \quad \hat{\sigma}_2^2 = .0221.$$

Thus the statistic

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{\sigma}_1^2 + \hat{\sigma}_2^2}} = -.3201$$

should be approximately an observation on a standard normal random variable if there are no differences in immune proportions between groups. In this case we conclude that the difference is not significant. Kersey et al. (1987, p. 465) come to a similar conclusion. Note that a crude ± 2 (estimated standard error) rule would not strongly suggest that p_1 and p_2 are strictly less than 1. But this crude test is invalid due to the boundary nature of the hypothesis.

Kersey et al. (1987, p. 465) also tested for differences between the autologous and allogeneic groups using the Mantel–Cox procedure, obtaining a non-significant difference (p value = .11). They did obtain significant differences between groups by comparing the KMEs for each group, using a confidence interval approach. However such an analysis leaves us with the problem of ascribing the difference between groups to a difference in immune proportions, or to a difference in relapse rates in early parts of the trial, or perhaps to both.

It would seem natural to suggest the use of the Cox proportional hazard model (Cox and Oakes (1984, Chapter 7)) for this kind of analysis. Unfortunately, if the data are even approximately distributed according to a mixture model as above, then the hazards of the two groups cannot be proportional, as mentioned by Kersey et al. (1987, p. 465). A recent paper by Kuk and Chen (1992) addresses this issue, and suggests a proportional conditional hazards model. It is however very computationally intensive and its theoretical properties are not yet known.

Finally, turning to the question of goodness of fit, the procedure suggested in Section 3 is simple to implement and seems a natural way to compare observed and fitted distributions. Its disadvantage is that the null distribution of the statistic will depend on the unknown censoring distribution. In some cases it may be possible to estimate this via the Kaplan–Meier procedure, but for the data in Section 3 there are too few censored observations to permit this. Other tests for goodness of fit to censored data have been proposed in the literature, for example that of Guilbaud (1988) based on a statistic of Kolmogorov–Smirnov type, but this procedure is more difficult to apply, requiring iteration to calculate the test statistic. In addition, with a sample size as small as 46, Kolmogorov–Smirnov type tests are known to have little power to detect moderate deviations from the null distribution. We have also checked the power of our suggested correlation procedure by some small simulations and found that, even for a moderate deviation from the null exponential distribution, such as a gamma distribution with shape parameter of 2, the values of r drop to below the 5th percentiles corresponding to the exponential null distribution, indicating a poor fit.

Finally we remark that model (1.1) is a special case of more general models in which further mixture components are included (see for example Larson and Dinse (1985) and Böhning et al. (1992)), but so far we have not attempted to program these models.

Acknowledgements

We are grateful to Prof. J. H. Kersey, University of Minnesota Cancer Center, for permission to use the leukemia data. Thanks also to Dr. Anne Goldman, Division of Biostatistics, University of Minnesota, for many helpful e-mail discussions and encouragement, and to Professor M. Aitkin and a number of referees for helpful comments.

Résumé

Dans un énchantillon de délais de survie censurés, la présence d'une proportion d'individus guéris qui ne sont exposés ni au décès, ni à l'échec, ni à la rechute, etc., peut étre signalée par l'observation d'un nombre relativement important de délais de censure longs. Des résultats théoriques récents sont résumés justifant une modélisation et des techniquies de tests semblables à celles habituellement utilisées pour de telles données. En particulier, on présente un test de "bornes" pour la présence de quéris dans une population, ainsi que des tests d'adéquation pour des distributions paramétriques des données. Les méthodes sont appliquées à des données de délais de rechute chez des patients atteints de leucémie.

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Received March 1993; revised October 1993 *and August* 1994; *accepted September* 1994.

Appendix

Here we will sketch a proof that, when the null hypothesis (2.3) is true, the asymptotic result (2.5) for the deviance holds. To keep the exposition simple, we will consider only the case that λ is known; extending this to the general case is technically complicated but similar in principle, and we leave the details to Zhou and Maller (1995). When λ is known, we can write the log-likelihood of a sample $t_1, \ldots, t_n$, apart from constants, as

$$\ell_n(p) = \sum_{i=1}^{n} \{c_i \log p + (1 - c_i)\log(1 - pF_i)\}, \quad 0 < p < 1,$$

where $F_i = 1 - e^{-\lambda t_i}$ and c_i are censor indicators; $c_i = 1$ if t_i is uncensored, and $c_i = 0$ otherwise. (See also Goldman's (1994) Appendix). Let $S_n(p) = d\ell_n(p)/dp$. It is not difficult to show that

$$X_n(p) = pS_n(p) = \sum_{i=1}^{n} \left(\frac{c_i - pF_i}{1 - pF_i}\right)$$

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is non-increasing in p on [0,1], and that $X_n(0) > 0$ with probability approaching 1 as $n \to \infty$. Thus there is a p' in (0,1) such that $X_n(p') = 0$, equivalently, $S_n(p') = 0$, if and only if $X_n(1) < 0$. Then

$$\tilde{p}_n = \begin{cases} p' & \text{if } X_n(1) < 0 \\ 1 & \text{if } X_n(1) \geq 0 \end{cases}$$

is the unique MLE of p in (0,1]. Suppose

$$E(e^{\lambda u}) = \int_{[0,\infty)} \frac{dG(y)}{1 - F(y)} = \sigma^2 + 1, \quad \text{say,}$$

is finite. By Lemma 2 of Ghitany, Maller, and Zhou (1994), $X_n(p)$ has mean zero and variance $n\sigma^2$. Since the t_i and c_i are i.i.d., we can apply the Central Limit Theorem to obtain, as $n \to \infty$,

$$\frac{X_n(1)}{\sqrt{n}} \xrightarrow{D} N(0, \sigma^2).$$

Thus $P[X_n(1) \geq 0] \to 1/2$, equivalently, $P[\tilde{p}_n = 1] \to 1/2$, and so $\tilde{p}_n$ is not asymptotically normal. However, using a Taylor expansion we can show, that for $x < 0$,

$$\begin{aligned} P\{\sigma\sqrt{n}(\tilde{p}_n - 1) \leq x | \tilde{p}_n < 1) &= P\{\sigma\sqrt{n}(\tilde{p}_n - 1) \leq x | X_n(1) < 0\} \\ &\to P\{N(0,1) \leq x | N(0,1) \leq 0\}, \end{aligned}$$

so $\sigma\sqrt{n}(\tilde{p}_n - 1)$ has a truncated normal distribution asymptotically. To test the null hypothesis H_0: $p = 1$, we use d_n as defined by (2.4), i.e.,

$$d_n = -2\{\ell_n(1) - \ell_n(\tilde{p}_n)\}.$$

When $\tilde{p}_n = 1$, $d_n = 0$, so

$$P\{d_n = 0\} \to \frac{1}{2}. \tag{A.1}$$

When $\tilde{p}_n < 1$, a Taylor expansion shows that

$$\begin{aligned} \ell_n(1) - \ell_n(\tilde{p}_n) &= (1 - \tilde{p}_n)S_n(\tilde{p}_n) + \frac{1}{2}(1 - \tilde{p}_n)^2 \frac{dS_n(p'_n)}{dp} \\ &= -\frac{1}{2}(1 - \tilde{p}_n)^2 F_n(p'_n), \end{aligned}$$

for some p'_n between $\tilde{p}_n$ and 1. Here $F_n(p) = -d^2\ell_n(p)/dp^2$ and we can use the weak law of large numbers to show that $F_n(p'_n)/n \xrightarrow{P} \sigma^2$. Thus, as $n \to \infty$,

$$P\{d_n^2 \leq x | \tilde{p}_n < 1\} \to P\{\chi_1^2 \leq x\}. \tag{A.2}$$

for $x > 0$. Equation (2.5) follows from (A.1)–(A.2) and the fact that $P[\tilde{p}_n = 1] \to 1/2$ as $n \to \infty$.

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